

Hybrid Quantum–Classical Control Simulator

Finite-Shot State Recovery, Hardware Limits, and Measurement-Aware Optimization

Purpose: Provide a compact, validated engineering simulator for parameterized two-qubit state preparation in which coherent calibration errors, open-system noise, finite-shot measurement, state inference, and classical retuning are treated in one end-to-end hybrid control loop.

Boris Kiefer

New Mexico State

University

bkiefer@nmsu.edu

[GitHub](#)

[LinkedIn](#)



Technical simulator • 8 October 2026 • Hybrid control, finite-shot inference, and hardware-recovery analysis.

Abstract

The simulator implements a validated hybrid quantum–classical control stack around a parameterized two-qubit state. The quantum layer evolves density matrices, supports explicit pre- and post-entangling noise locations, and includes amplitude damping, pure dephasing, and depolarizing channels in Kraus form. The control state adds an independently tunable relative phase, so entanglement alone does not uniquely identify the target. The measurement layer replaces privileged access to the exact density matrix with finite-shot Pauli measurements, linear state reconstruction, and physical projection. A classical optimizer then retunes control parameters from estimated concurrence and target-state fidelity. Systematic control offsets are separated from irreversible decoherence by comparing nominal performance, finite-shot recovery, and an exact noisy-hardware ceiling. For the current circuit/noise family, X-state structure reduces the inferred state from 15 nontrivial Pauli observables to a seven-observable model-aware set.

Derived impact statement

Impact hypothesis. A hybrid quantum–classical simulator is most useful as an engineering scaffold when it distinguishes what feedback can recover from what the hardware cannot deliver. The present implementation makes that distinction explicit: coherent calibration errors are retunable, decoherence imposes a physical ceiling, finite-shot measurements limit inference, and prior structure in the reachable state family can reduce the measurement burden.

Executive decision frame

Control	Can a classical feedback layer recover systematic control offsets using only finite-shot measurement-derived state estimates rather than direct access to the simulated density matrix?	Hardware	What fraction of state degradation is recoverable by retuning, what fraction is irreducible under the selected noise model, and what measurement resources are required to approach the physical ceiling?
----------------	---	-----------------	---

Customer discovery

Quantum hardware teams	Control / calibration teams	Systems architecture	Platform-specific extensions
Map device parameters and calibrated error channels into a compact state-preparation model; identify which errors are coherent, stochastic, leakage-like, or irreversible.	Evaluate whether finite-shot feedback can compensate drift or calibration offsets and quantify controller performance relative to a device-specific physical ceiling.	Use observable count, shot count, optimizer evaluations, and model assumptions to expose quantum/classical resource tradeoffs rather than treating the backend as an opaque cost function.	Replace the current generic two-qubit backend with modality-specific physics: gate-based DV hardware, photonic loss/erasure and interferometers, sensing channels, or experimentally fitted noise maps.

1. Scope and engineering question

The simulator is a deliberately small but complete hybrid control model. Its purpose is not to emulate a particular commercial processor. Instead, it provides a traceable systems scaffold in which the quantum backend, measurement process, classical inference, optimizer, and physical performance limit are separately visible and independently testable. This architecture follows the general parameterized-circuit/classical-optimizer pattern used in hybrid and variational quantum algorithms, while shifting the emphasis from algorithmic energy minimization to state preparation, calibration, and recoverability [6].

The present backend prepares a two-qubit state from two controls, α and ϕ , evolves a density matrix through user-selected noise, generates finite-shot Pauli data, reconstructs an estimated state $\hat{\rho}$, and returns estimated concurrence and target-state fidelity to a classical controller. Exact density-matrix quantities are retained only as simulator diagnostics and to compute a noisy-hardware oracle ceiling; they are not used by the finite-shot controller.

Core control loop:

$$(\alpha, \phi)_{\text{cmd}} \longrightarrow \rho_{\text{hardware}} \longrightarrow \text{finite-shot Pauli records} \longrightarrow \hat{\rho} \longrightarrow (\hat{C}, \hat{F}) \longrightarrow \text{classical retuning.}$$

2. Ideal state, target specification, and observability

Starting from $|00\rangle$, the ideal circuit applies $R_y(\alpha)$ to qubit 0, a CNOT with qubit 0 as control, and a phase gate $P(\phi) = \text{diag}(1, e^{i\phi})$ to qubit 0. The resulting pure state is

$$|\psi(\alpha, \phi)\rangle = \cos \frac{\alpha}{2} |00\rangle + e^{i\phi} \sin \frac{\alpha}{2} |11\rangle. \quad (1)$$

The target is specified by a concurrence $C_t \in [0, 1]$ and target phase ϕ_t . On the principal branch,

$$\alpha_t = \arcsin C_t, \quad |\psi_t\rangle = |\psi(\alpha_t, \phi_t)\rangle. \quad (2)$$

For the ideal family in Eq. (1), the concurrence is

$$C(\alpha, \phi) = |\sin \alpha|, \quad (3)$$

which is independent of ϕ . Consequently, concurrence alone leaves a flat control direction. The state fidelity resolves that missing information. For a pure target and a possibly mixed realized state,

$$F = \langle \psi_t | \rho | \psi_t \rangle. \quad (4)$$

For two ideal states from Eq. (1),

$$F = \left| \cos \frac{\alpha_t}{2} \cos \frac{\alpha}{2} + e^{i(\phi - \phi_t)} \sin \frac{\alpha_t}{2} \sin \frac{\alpha}{2} \right|^2. \quad (5)$$

At $\alpha = \alpha_t = \pi/2$, $|\Phi^+\rangle$ and $|\Phi^-\rangle$ both have $C = 1$ but zero mutual fidelity. This is the simulator's simplest explicit demonstration that an entanglement metric is not a complete state-control objective.

3. Density-matrix backend and open-system channels

The V2 backend uses density matrices throughout. Unitary evolution is

$$\rho \longrightarrow U \rho U^\dagger, \quad (6)$$

and a quantum channel in Kraus form is

$$\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger, \quad \sum_k K_k^\dagger K_k = I, \quad (7)$$

which guarantees a completely positive, trace-preserving map when the completeness condition is satisfied [1, 2].

The circuit exposes two explicit noise locations,

$$|00\rangle \xrightarrow{R_y(\alpha)} \mathcal{N}_A \xrightarrow{\text{CNOT}} \xrightarrow{P(\phi)} \mathcal{N}_B \longrightarrow \rho_{\text{out}},$$

so identical channel strengths can have different consequences depending on when they act. For example, amplitude damping on qubit 1 before the CNOT is irrelevant because that qubit is still in $|0\rangle$, whereas the same damping after entanglement formation reduces both concurrence and fidelity.

Amplitude damping

With relaxation probability γ ,

$$K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}. \quad (8)$$

For one-sided damping of $|\Phi^+\rangle$, the implementation reproduces the analytic result

$$C = \sqrt{1-\gamma}. \quad (9)$$

Pure dephasing

The convention used in the simulator multiplies one-qubit coherence by $1-p$ while leaving populations unchanged:

$$K_0 = \sqrt{1-p} I, \quad K_1 = \sqrt{p} |0\rangle\langle 0|, \quad K_2 = \sqrt{p} |1\rangle\langle 1|. \quad (10)$$

For one-sided dephasing of $|\Phi^+\rangle$,

$$C = 1-p, \quad F = 1 - \frac{p}{2}. \quad (11)$$

Depolarizing channel

The local convention is

$$\mathcal{E}_{\text{dep}}(\rho) = (1-p)\rho + p\frac{I}{2}, \quad (12)$$

implemented by

$$K_0 = \sqrt{1 - \frac{3p}{4}} I, \quad K_j = \sqrt{\frac{p}{4}} \sigma_j, \quad j \in \{x, y, z\}. \quad (13)$$

For one-sided depolarization of $|\Phi^+\rangle$,

$$C = \max\left(0, 1 - \frac{3p}{2}\right), \quad F = 1 - \frac{3p}{4}. \quad (14)$$

When multiple channels are combined, the simulator applies amplitude damping, then dephasing, then depolarization. The order is explicit because general channels need not commute.

4. Mixed-state concurrence and target-state fidelity

Mixed-state concurrence is evaluated with the Wootters construction [3]. Define

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y) \rho^* (\sigma_y \otimes \sigma_y). \quad (15)$$

Let λ_i be the decreasing square roots of the eigenvalues of $\rho \tilde{\rho}$. Then

$$C(\rho) = \max(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4). \quad (16)$$

The controller uses both C and the pure-target fidelity in Eq. (4). In the ideal circuit a concurrence-only objective identifies α but cannot identify ϕ ; a state-aware objective closes that observability gap.

5. Finite-shot Pauli inference

Direct access to ρ is removed from the control loop. For a Pauli observable P with exact expectation $m = \text{Tr}(\rho P)$, the ± 1 outcome probabilities are

$$p_{\pm} = \frac{1 \pm m}{2}. \quad (17)$$

Finite-shot outcomes are sampled binomially, giving an estimator

$$\hat{m} = \frac{n_+ - n_-}{N_{\text{shots}}}. \quad (18)$$

For full two-qubit Pauli reconstruction, the simulator estimates the 15 nontrivial Pauli products and performs linear inversion,

$$\hat{\rho}_{\text{raw}} = \frac{1}{4} \sum_{\mu, \nu \in \{I, X, Y, Z\}} \langle \widehat{\sigma_{\mu} \otimes \sigma_{\nu}} \rangle \sigma_{\mu} \otimes \sigma_{\nu}, \quad (19)$$

with $\langle II \rangle = 1$. Linear-inversion tomography is standard, while statistically consistent reconstructions are often obtained with constrained or maximum-likelihood methods [4, 5]. The current simulator uses a deliberately transparent physicalization step: Hermitize $\hat{\rho}_{\text{raw}}$, diagonalize, clip negative eigenvalues to zero, and renormalize the trace. This is not claimed to be maximum-likelihood tomography.

The finite-shot controller sees only $\hat{C} = C(\hat{\rho})$ and $\hat{F} = \langle \psi_t | \hat{\rho} | \psi_t \rangle$. Exact C and F are reported separately as simulator diagnostics.

6. Hybrid control objective and search

The state-aware control objective is

$$\mathcal{L} = w_C \left(\hat{C} - C_t \right)^2 + w_F \left(1 - \hat{F} \right), \quad (20)$$

with non-negative w_C and positive w_F . The baseline optimizer is intentionally simple: a global coarse grid in (α, ϕ) followed by local grid refinement around the best finite-shot candidate. Each candidate receives an independent finite-shot sample. This derivative-free search is transparent, deterministic under a fixed random seed, and makes total measurement cost explicit. It is a baseline controller, not a claim of sample-optimal optimization.

A separate exact optimizer has access to the simulated density matrix and is used only to compute the noisy-hardware ceiling. It therefore acts as an oracle diagnostic rather than an operational controller.

7. Hardware offsets and recoverability decomposition

Systematic calibration errors are introduced as command-to-hardware offsets,

$$\alpha_{\text{actual}} = \alpha_{\text{cmd}} + \delta\alpha, \quad \phi_{\text{actual}} = \phi_{\text{cmd}} + \delta\phi. \quad (21)$$

These coherent errors are, in principle, recoverable by retuning. Incoherent channels such as dephasing impose a ceiling that retuning cannot exceed.

The simulator reports four fidelity levels:

$$F_{\text{ideal}} = 1, \quad (22)$$

$$F_{\text{nominal}} = \text{hardware result from uncorrected ideal commands}, \quad (23)$$

$$F_{\text{controlled}} = \text{true state produced by the finite-shot-selected commands}, \quad (24)$$

$$F_{\text{ceiling}} = \text{best exact fidelity supported by the noisy hardware model}. \quad (25)$$

The resulting decomposition is

$$\Delta F_{\text{recoverable}} = F_{\text{ceiling}} - F_{\text{nominal}}, \quad (26)$$

$$\Delta F_{\text{recovered}} = F_{\text{controlled}} - F_{\text{nominal}}, \quad (27)$$

$$\Delta F_{\text{controller}} = F_{\text{ceiling}} - F_{\text{controlled}}, \quad (28)$$

$$\Delta F_{\text{irreducible}} = F_{\text{ideal}} - F_{\text{ceiling}}. \quad (29)$$

By construction,

$$F_{\text{ideal}} - F_{\text{nominal}} = \Delta F_{\text{recoverable}} + \Delta F_{\text{irreducible}}. \quad (30)$$

This is the central engineering distinction in the simulator: calibration/feedback opportunity is reported separately from the physical noise floor.

8. Model-aware X-state measurement reduction

For the present circuit and implemented channel family, the state remains in the two-qubit X-state class,

$$\rho_X = \begin{pmatrix} a & 0 & 0 & z \\ 0 & b & w & 0 \\ 0 & w^* & c & 0 \\ z^* & 0 & 0 & d \end{pmatrix}. \quad (31)$$

This structure is verified numerically after combinations of pre- and post-entangling amplitude damping, dephasing, and depolarization. The complete X state is reconstructed from only

$$\boxed{ZI, IZ, ZZ, XX, YY, XY, YX}. \quad (32)$$

The populations follow from

$$a = \frac{1 + \langle ZI \rangle + \langle IZ \rangle + \langle ZZ \rangle}{4}, \quad (33)$$

$$b = \frac{1 + \langle ZI \rangle - \langle IZ \rangle - \langle ZZ \rangle}{4}, \quad (34)$$

$$c = \frac{1 - \langle ZI \rangle + \langle IZ \rangle - \langle ZZ \rangle}{4}, \quad (35)$$

$$d = \frac{1 - \langle ZI \rangle - \langle IZ \rangle + \langle ZZ \rangle}{4}, \quad (36)$$

and the coherences from

$$\Re z = \frac{\langle XX \rangle - \langle YY \rangle}{4}, \quad \Im z = -\frac{\langle XY \rangle + \langle YX \rangle}{4}, \quad (37)$$

$$\Re w = \frac{\langle XX \rangle + \langle YY \rangle}{4}, \quad \Im w = \frac{\langle XY \rangle - \langle YX \rangle}{4}. \quad (38)$$

At equal shots per independently sampled observable, this changes the current simulator accounting from 15 to 7 observables per candidate, a reduction of

$$1 - \frac{7}{15} = 53.3\%. \quad (39)$$

For the frozen 659-evaluation benchmark at 2000 shots per observable, the synthetic count decreases from 19,770,000 to 9,226,000 observable-shots.

Measurement-accounting caveat. The current simulator counts each Pauli observable independently. On hardware, one product-basis measurement setting can estimate several compatible Pauli expectations from the same records. Therefore 15-to-7 is an observable-count reduction under the simulator's current accounting, not a claim that experimental acquisition time must fall by exactly 53.3%. A setting-aware sampler is a natural next extension.

9. Verification gates

The implementation was developed incrementally, with each layer frozen only after its analytic or regression tests passed. The verification table below summarizes the verified stack.

Cell	Added capability	Primary verification gate
1	Density-matrix core	Reproduces the ideal pure-state circuit, purity, concurrence $ \sin \alpha $, and unit target fidelity.
2	CPTP channel library	Kraus completeness, zero-noise identity, positivity, and analytic Bell-state concurrence under damping, dephasing, and depolarization.
3	Explicit noise locations	Analytic post-entangling C and F ; pre-entangling noise on an inactive target qubit can be irrelevant.
4	Relative phase control	Concurrence independent of ϕ ; general fidelity formula; $ \Phi^+\rangle$ and $ \Phi^-\rangle$ have equal concurrence and zero mutual fidelity.
5	Exact hybrid optimization	Concurrence-only control cannot identify phase; state-aware control recovers both α and ϕ .
6	Finite-shot tomography	Exact Pauli reconstruction of pure/mixed states; reproducible shot sampling; physical reconstructed state after PSD projection.
7	Finite-shot hybrid loop	Controller objective uses estimated metrics only; fixed-seed search converges near target; shot budget tracked explicitly.
8	Hardware offsets / ceiling	Coherent offsets are recoverable; dephasing ceiling reproduced analytically; recoverable and irreducible fidelity gaps close.
9	X-state compression	X-state preservation verified; seven observables exactly reconstruct the model family; 7/15 observable accounting reproduced.
10	Integrated dashboard	Frozen benchmark, physical ceiling, compressed measurement backend, and recovery metrics reproduced end-to-end.

10. Technical limitations

The present simulator is intentionally bounded. These boundaries should be treated as explicit interface points rather than hidden approximations.

- The backend contains two qubits and one parameterized Bell-like state family; scaling behavior is not inferred from this small model.
- The implemented channels are phenomenological CPTP maps. They are not yet parameterized by device timing, T_1/T_2 , gate duration, crosstalk, drift spectra, or experimentally fitted process matrices.
- The seven-observable reduction is valid only while the reachable/noisy state remains in the X-state family. Any extension that generates off-X coherences must trigger the leakage diagnostic and expand the measurement model.
- The physicalization of finite-shot linear inversion uses eigenvalue clipping and trace renormalization. It is transparent but is not maximum-likelihood or Bayesian tomography [4, 5].
- The coarse-to-fine controller is a transparent baseline, not a sample-optimal optimizer. It also does not yet allocate shots adaptively across candidates or observables.
- The simulator does not include state-preparation-and-measurement (SPAM) error, detector bias, readout confusion matrices, leakage outside the computational subspace, non-Markovian noise, or feedback latency.
- For photonic dual-rail encodings, propagation or coupling loss should generally be modeled as erasure/leakage out of the logical subspace rather than identified directly with qubit amplitude damping. A photonic backend should add the appropriate vacuum/erasure sector explicitly.
- The exact hardware ceiling is a simulator oracle used for diagnosis. A physical system would require independent characterization or bounds to estimate the corresponding ceiling experimentally.

11. Test case: finite-shot recovery under calibration error and dephasing

The following case is the frozen compressed-measurement benchmark used by the integrated dashboard.

Input	Value
Target concurrence	$C_t = 1.000000$
Target phase	$\phi_t = 0$
Systematic control offsets	$\delta\alpha = +0.25$ rad, $\delta\phi = +0.70$ rad
Post-control dephasing	$p_0 = 0.20$, $p_1 = 0$
Measurement model	Seven-observable X-state reconstruction, Eq. (32)
Shots per observable	2000
Random seed	3
Search	17×25 global grid plus two refinement rounds
Evaluations	659

The ideal Bell target has $\alpha_t = \pi/2$. Exact compensation of the coherent offsets would therefore command

$$\alpha_{\text{cmd}}^{(0)} = \frac{\pi}{2} - 0.25 = 1.320796, \quad \phi_{\text{cmd}}^{(0)} = -0.700000. \quad (40)$$

One-sided dephasing at $p = 0.20$ sets the analytic physical ceiling from Eq. (11),

$$C_{\text{ceiling}} = 0.800000, \quad F_{\text{ceiling}} = 0.900000. \quad (41)$$

Output	Estimated / selected	True simulator diagnostic
Nominal concurrence	–	0.775130
Nominal fidelity	–	0.796426
Selected α_{cmd}	1.349903	–
Selected ϕ_{cmd}	–0.753982	–
Controlled concurrence	0.805817	0.799661
Controlled fidelity	0.902702	0.899248
Physical ceiling	–	$C = 0.800000$, $F = 0.900000$
Synthetic measurement budget	9,226,000 observable-shots	
Full-tomography comparison	19,770,000 observable-shots	

The finite-shot estimate can fluctuate slightly above the exact physical ceiling; the true simulator diagnostic remains below it. In this realization, the selected controls recover the nominal fidelity from 0.796426 to 0.899248, leaving a true controller gap of approximately 7.52×10^{-4} below the $F = 0.9$ hardware ceiling. The remaining 0.1 gap to unit fidelity is irreducible within the specified one-sided dephasing model.

12. Outlook and extension path

The present V2 implementation is a validated warm-start architecture. The next extensions should be driven by a target hardware stack rather than by adding generic complexity.

- **Hardware-calibrated channels:** map measured gate errors, T_1/T_2 , gate times, drift, and correlated errors into the quantum backend; allow gate-dependent and time-dependent channels.
- **Setting-aware and adaptive measurement:** replace independent-observable shot accounting with product-basis measurement settings, reuse compatible outcomes, allocate shots adaptively, and compare direct objective estimation against full state reconstruction.
- **Advanced estimators:** add maximum-likelihood or Bayesian state estimation, uncertainty propagation, and confidence intervals on C , F , and the inferred recovery fraction.
- **Online calibration and drift tracking:** replace static offsets with time-dependent parameters and close the loop with sequential estimation or filtering.
- **Alternative classical controllers:** benchmark SPSSA, Bayesian optimization, covariance-matrix strategies, and model-predictive control against the transparent coarse-to-fine baseline using common shot budgets.
- **Leakage / erasure sectors:** extend the Hilbert space to model non-computational states, photonic vacuum/erasure, detector loss, and heralding; retain the current two-qubit model as the zero-leakage validation limit.

- **Beyond two qubits:** replace full density-matrix reconstruction with reduced-observable, shadow, tensor-network, or model-specific estimators as state dimension grows; retain explicit resource accounting.
- **Backend abstraction:** preserve the classical inference/control interface while replacing the quantum block with gate-based DV hardware, photonic circuits, sensing models, or experimentally supplied process maps.

The architectural invariant is the separation of *command*, *quantum evolution*, *measurement*, *classical inference*, *control update*, and *physical ceiling*. That separation makes the simulator useful as a technical translation layer: once the target hardware and error mechanisms are known, the idealized backend can be replaced incrementally without changing the end-to-end control logic.

References

References

- [1] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, 10th Anniversary ed. (Cambridge University Press, Cambridge, 2010), ISBN 978-1-107-00217-3.
- [2] H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, Oxford, 2002). doi:10.1093/acprof:oso/9780199213900.001.0001.
- [3] W. K. Wootters, “Entanglement of Formation of an Arbitrary State of Two Qubits,” *Phys. Rev. Lett.* **80**, 2245–2248 (1998). doi:10.1103/PhysRevLett.80.2245.
- [4] D. F. V. James, P. G. Kwiat, W. J. Munro, and A. G. White, “Measurement of qubits,” *Phys. Rev. A* **64**, 052312 (2001). doi:10.1103/PhysRevA.64.052312.
- [5] J. B. Altepeter, E. R. Jeffrey, and P. G. Kwiat, “Photonic State Tomography,” *Advances in Atomic, Molecular and Optical Physics* **52**, 105–159 (2005). doi:10.1016/S1049-250X(05)52003-2.
- [6] M. Cerezo, A. Arrasmith, R. Babbush, *et al.*, “Variational quantum algorithms,” *Nature Reviews Physics* **3**, 625–644 (2021). doi:10.1038/s42254-021-00348-9.